BRSM Project Report - Sentence Memorability

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Abstract:

This report tells about the complete statistical analysis pipeline for the Sentence Memorability experiment, examining how noun memorability, sentence voice, reaction time patterns, and cognitive fatigue influence recognition memory across 112 participants. Using a continuous recognition paradigm, participants viewed sentences constructed from four word-memorability conditions (HH, HL, LH, LL) crossed with Active and Passive voice, by generating Corrected Memorability Scores (CMS) and Word Recognition (WR) accuracy as primary dependent variables. Eight formal hypotheses were tested using appropriate within-subjects non-parametric methods. Key findings include: (1) sentences with two low-memorability nouns (LL) exhibited significantly lower CMS than all other conditions (Friedman $\chi^2(3)=24.26$, $p<.001$); (2) grammatical voice did not affect recognition accuracy but did significantly slow retrieval speed (H4: W=2193, p=.002, r=.33); (3) trial-level reaction time predicted accuracy in the opposite direction to the classic speed-accuracy trade-off, faster responses were significantly more accurate (GEE OR=0.45, $p<.001$); (4) performance improved rather than declined across blocks, indicating a learning effect rather than cognitive fatigue; (5) syntactic transformation of sentence voice produced a substantial 12.8% accuracy penalty in word-recognition judgements (H7: W=4521, $p<.001$, r=.54); and (6) within-subject RT fluctuations predicted wording-retrieval accuracy, supporting the Memory Strength account (H8). Findings are interpreted through Dual Coding Theory (Paivio, 1971) and psycholinguistic accounts of passive processing (Ferreira, 2003).

1. Introduction:

Memory for sentences varies systematically: some are more memorable due to features like concreteness, imagery, and emotional richness of words, while abstract, less meaningful words tend to be forgotten more easily. Building on prior findings at the word level, this study explores how such memorability extends to entire sentences.

Using a continuous recognition task, participants identified repeated sentences (Item Recognition) and then judged word familiarity (Word Recognition). Sentences followed an S–V–O structure and varied by word memorability type (HH, HL, LH, LL) and voice (active/passive), creating eight conditions. Performance was measured using Corrected Memorability Score and WR accuracy. The study tests eight hypotheses related to memorability, syntax, and cognitive factors, using appropriate within-subject statistical methods for repeated measures.

2. Methodology:

2.1: Experimental Design:

This study used a continuous recognition paradigm. Participants viewed sentences and indicated whether each sentence had been previously presented by pressing the spacebar (IR response). Immediately following an IR press, a Yes/No WR prompt appeared. Sentences were constructed in S–V–O format with verbs held at medium concreteness to isolate noun-level memorability effects. The experiment comprised a practice phase followed by three experimental blocks separated by rest breaks.

Word Memorability was manipulated at four levels (HH, HL, LH, LL) crossed with Voice (Active, Passive), producing 8 experimental conditions. High-Frequency filler sentences (HF) were included to provide a reliable FA baseline. Approximately 10 validation sentences per block served as attention checks.

2.2 Participants:

A total of 334 participants were included in the experiment (according to the text file provided with the dataset). But we are performing the analysis on only part of them [232 - 398] which is 114 and we removed some of the participants' data (2 participants). So there are 112. Participants were excluded based on the validation criterion:

- Correct IR > (Wrong IR / 2) + Missed IR (assuming already done before)

- All Blocks that failed this validation test were excluded from memorability score calculation. (assuming that this is already done)

2.3 Variables:

Independent Variables:

- Word Memorability Type (HH, HL, LH, LL)
  (Note: Raw stimulus IDs use HVL, LVH, and LVL; these are relabelled as HL, LH, and LL respectively throughout this report)
- Voice (Active, Passive)

Dependent Variable:

- Corrected Memorability Score (Primary)
- WR Accuracy, IR Reaction time, WR reaction time (Secondary)

Corrected memorability and WR Accuracy was calculated as:

- Corrected Memorability = Hit Rate − False Alarm Rate
- WR Accuracy = Correct WRs / Total WRs

Hit Rate is the proportion of repeated target sentences in a condition that the participant correctly identified (Hits ÷ (Hits + Misses)). False Alarm Rate is the proportion of first-presentation sentences on which the participant incorrectly pressed the spacebar (which means the sentence is shown for the first time but participant marks it as seen), calculated globally across all non repeat, non-validation presentations and subtracted uniformly from every condition's hit rate

2.4 Block Validation and Data Processing:

Each session was divided into three blocks separated by rest breaks. To screen for inattentive participants, each block contained validation sentences (Nearly 10). The dataset has a Validation sentence column where true represents the validation sentence. A block was retained only if the participant passed the following criterion:

Correct Validation IRs > (Wrong Validation IRs ÷ 2) + Missed Validation IRs

Correct = correctly identified, Wrong = Incorrectly identified, Missed = did not respond at all.

Processing steps applied to the raw log data:

- Practice phase rows and timing bookkeeping rows were removed for all the users.
- Sessions were splitted into blocks based on Rest Phase Row.
- The attention-check criteria were applied to each block.
- For each block, if its valid, target-type repeat sentences were classified as Hits or Misses.
- False alarms were counted from non-repeat presentations.
- Corrected memorability was computed per condition. The final analysis dataset contained 896 records (112 participants × 8 conditions), with N = 224 observations per condition.

2.5 Secondary Data Pipelines for H5 and H6:

Pipeline 1 : Trial-Level Data for H5 (reactions_data.csv):

Every event row in the raw logs that contained a valid Reaction_time_IR value was extracted. This includes both Hits (correct recognitions) and False Alarms (incorrect presses on new sentences), the only trial types that produce an IR response and therefore a valid RT. Misses and Correct Rejections were excluded as they produce no RT. Each extracted row records: *Participant ID (PID), Trial ID (TID, incrementing per participant), current Block number (tracked by counting Rest Phase started events), IR Accuracy (boolean: 1 = correct, 0 = false alarm), and IR Reaction Time (ms)*. Practice-phase rows were excluded. The resulting file contains approximately 15,170 trial-level rows across 112 participants.

Pipeline 2 : Block-Level Aggregates for H6 (block_accuracy_data.csv):

From the same trial-level extraction above, reactions were aggregated per participant per block. For each (PID, Block) pair, two aggregates were computed: (1) Reaction Accuracy = Total Hits / (Total Hits + Total False Alarms), using WR accuracy to classify hits and false alarms when available, or IR accuracy as a fallback for trials without a WR probe (e.g., filler sentences); (2) Mean RT = arithmetic mean of all valid IR reaction times in that block. This yields one row per participant per block, producing approximately 342 block-level records (114 participants × 3 blocks).

2.6 Data Processing for H7 and H8:

To assess wording retrieval and memory strength, data were filtered to include only correctly recognized main-experiment target repeats (Item-Recognition[IR] Hits). For H7, the true experimental condition

(Same vs. Transformed) was derived by combining participants' Word-Recognition (WR) responses with their binary accuracy. For H8, Item-Recognition reaction times (RT_IR) were restricted to a standard 200–4000 ms window to capture genuine cognitive processing and exclude physiological fast-guesses or inattentive outliers (consistent with Whelan). Cleaned RTs were then within-subject standardized (z-scored) to isolate individual trial-level fluctuations, removing baseline speed and variance differences.

3. Statistical Approach and Normality Testing:

Before selecting inferential tests, all dependent variables were tested for normality using the Shapiro-Wilk

3.1 Normality of CMS:

All eight word-type × voice conditions showed significantly non-normal CMS distributions (all $p < .001$). The distributions appear left-skewed reflecting ceiling effects in recognition performance. QQ-plot inspection confirms systematic deviation at both tails.

Critical methodological correction from Report 1: Report 1 used Kruskal-Wallis (between-subjects) and Mann-Whitney U (independent samples). Because the same 112 participants contribute data in ALL 8 conditions, the design is fully within-subjects. The correct non-parametric equivalents are: (1) Friedman test replaces Kruskal-Wallis for the word-type omnibus effect; (2) paired Wilcoxon Signed-Rank replaces Mann-Whitney U for the voice comparison.

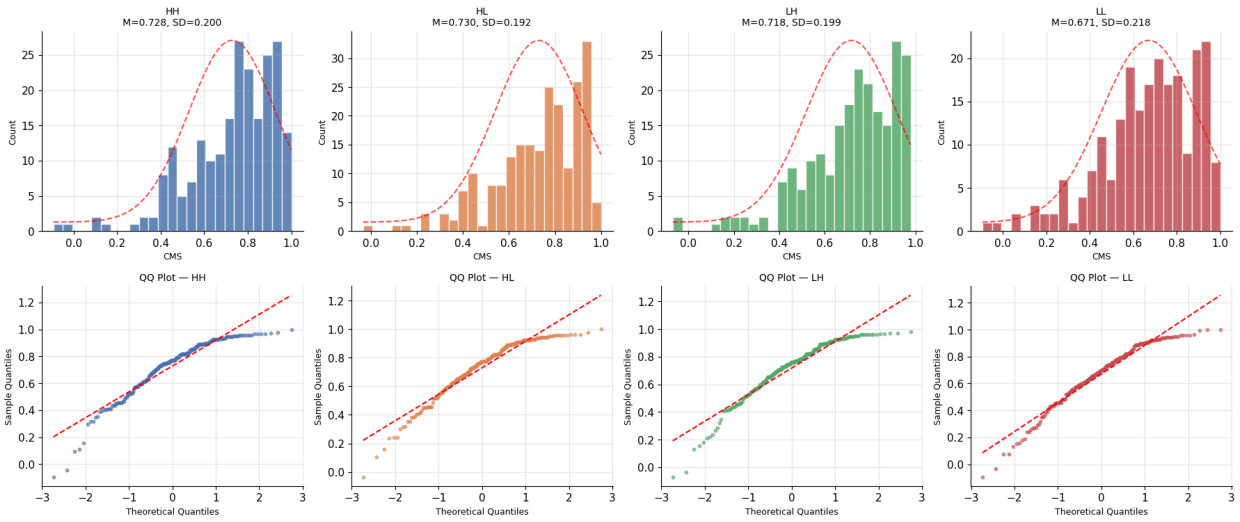


Fig: Corrected Memorability Score Distribution per Word Type - Histogram+KDE and QQ Plots

3.2 Normality of Trial Level RT:

The trial-level IR Reaction Time data ($N \approx 15{,}170$ trials from reactions_data.csv) was tested for normality. The Shapiro-Wilk test confirmed strong non-normality: IR Reaction Time (trial-level): $W = 0.9206$, $p = 3.58 \times 10^{-66}$ => Non-Normal (right-skewed)

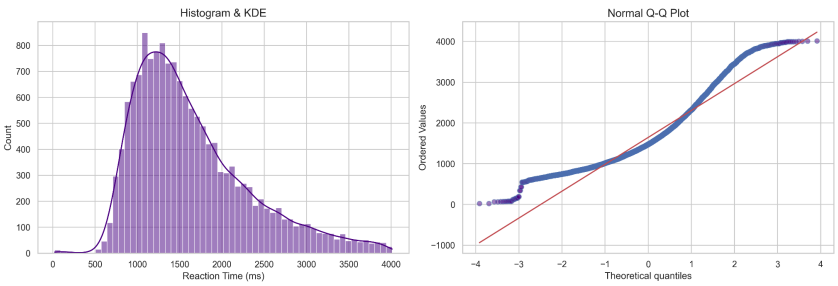


Fig: Distribution of Reaction Time for normality check : Histogram+KDE and QQ Plot

3.3 Normality of Block Level Variables:

The block-level aggregates ($N \approx 342$ rows from block_accuracy_data.csv) were tested separately for each variable:

- Block Mean RT: $W = 0.9911$, $p = 3.61 \times 10^{-2}$ => Non-Normal
- Block Accuracy: $W = 0.9673$, $p = 5.88 \times 10^{-7}$ => Non-Normal

Although the W statistics are higher (closer to 1.0) than the trial-level RT - reflecting the smoothing effect of aggregation. Both variables still reject normality at $\alpha = .05$. Notably, Shapiro-Wilk is sensitive to sample size: even minor deviations are flagged as significant with larger N.

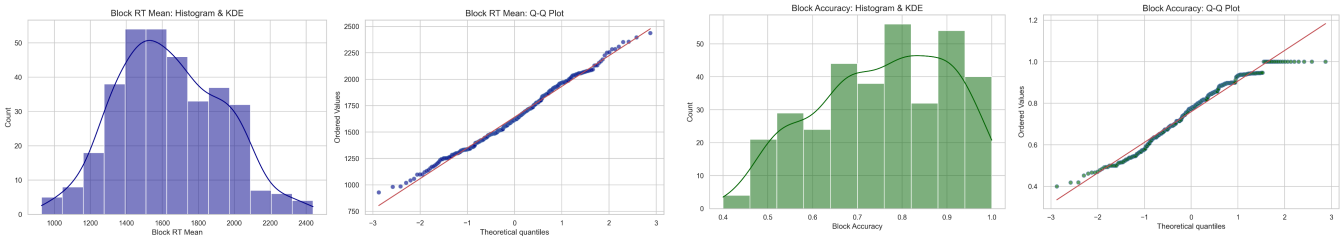
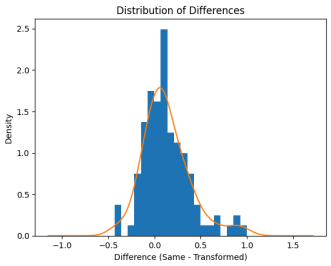


Fig: Distribution of RT Mean and Block Accuracy : Histogram + KDE and QQ Plots

3.4 Normality of Paired WR Accuracy Differences:

The paired differences in WR accuracy between the Same and Transformed conditions were tested. The Shapiro-Wilk test confirmed a significant deviation from normality ($p = 0.0001$), justifying the use of the non-parametric Wilcoxon Signed-Rank test.



For the memory strength model, the dependent variable is binary (WR Accuracy: Correct/Incorrect), making normality testing inapplicable and necessitating a Generalized Estimating Equations (GEE) approach.

Collectively, the consistent violation of normality across all continuous measures justifies the use of non-parametric tests (Friedman, Wilcoxon Signed-Rank) and appropriate generalized models throughout H1–H8.

4. Hypothesis Testing and Results:

4.1 Effect of Word Memorability Type of CMS:

4.1.1: Hypothesis Statement:

[H_0]: There is no significant difference in Corrected Memorability Score (CMS) across the four word memorability conditions (HH, HL, LH, LL).

[H_a] : Word memorability type significantly affects CMS, such that sentences composed entirely of low-memorability nouns (LL) will have lower CMS than sentences containing at least one high-memorability noun (HH, HL, LH).

4.1.2: Rationale:

High-memorability words are typically more concrete, imageable, and emotionally rich (Rubin & Friendly, 1986; Paivio, 1971), making them more distinctive and easier to encode and recognise. Dual Coding Theory (Paivio, 1971) predicts that high-memorability nouns activate both verbal and imagistic memory representations, producing richer encoding. LL sentences lack this dual-trace advantage at both noun positions, predicting reliably lower recognition.

4.1.3 Statistical Test: Friedman Test + Wilcoxon Post-hoc

Data were pivoted to participant × word-type format (each participant contributing one CMS value per condition, averaged across voice). The Friedman test was applied as the non-parametric equivalent of a one-way repeated-measures ANOVA. Post-hoc pairwise Wilcoxon Signed-Rank tests with Holm correction were used to identify which specific pairs differed. Effect sizes were computed as Kendall's W (omnibus) and rank-biserial correlation r (per pair).

4.1.4 Descriptive Statistics

CMS by word memorability type (N=112 per condition, median = primary statistic due to non-normality):

Condition	N	Median CMS	Mean CMS	SD	IQR	Skewness
HH (High–High)	112	0.7743	0.728	0.194	0.198	−1.21
HL (High–Low)	112	0.7535	0.730	0.191	0.202	−1.18

Condition	N	Median CMS	Mean CMS	SD	IQR	Skewness
LH (Low–High)	112	0.7569	0.718	0.196	0.201	−1.19
LL (Low–Low)	112	0.6875	0.671	0.202	0.213	−1.05

4.1.5 Inferential Statistics:

The Friedman test revealed a significant overall effect of word type on CMS:

$$\chi^2(3, N=112) = 24.26, p < .001, \text{Kendall's } W = .07 \text{ (small effect)}$$

Post-hoc Wilcoxon tests with Holm correction showed that LL produced significantly lower CMS.

Pair	Median A	Median B	W	p_raw	p_holm	Significant	r_rb	Effect
HH vs HL	0.7743	0.7535	2884.0	0.3254	0.3254	no	0.11	small
HH vs LH	0.7743	0.7569	2761.5	0.5612	0.5612	no	0.07	negligible
HH vs LL	0.7743	0.6875	2008.0	0.0005	0.0015	yes	0.45	medium
HL vs LH	0.7535	0.7569	2913.0	0.9742	0.9742	no	−0.01	negligible
HL vs LL	0.7535	0.6875	1736.5	<.0001	<.0001	yes	0.56	large
LH vs LL	0.7569	0.6875	2128.0	0.0026	0.0078	yes	0.38	medium

No other pairwise differences (HH vs HL, HH vs LH, HL vs LH) were statistically significant after correction, indicating that the memorability deficit was specific to the LL condition

4.1.6: Visualizations:

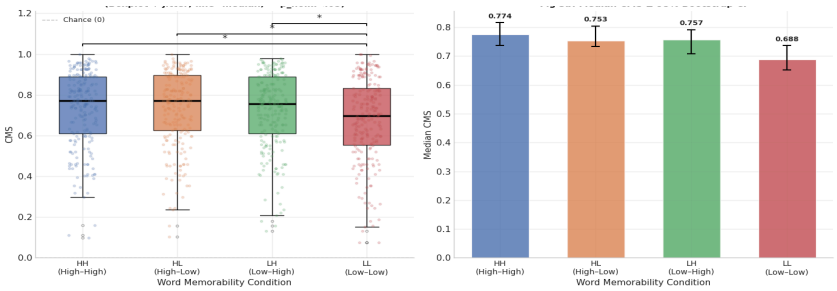


*Fig: [left] CMS by word type (Boxplot + Jitter, Line = Mean, *=p_holm<0.5) [right] Median CMS+=95% CI*

4.1.7: Conclusion:

H_0 is rejected. Word memorability type significantly affects CMS. The LL condition produced reliably lower memorability scores than all conditions containing at least one high-memorability noun. The median CMS pattern (HH: 0.774 > LH: 0.757 ≈ HL: 0.754 > LL: 0.688) suggests that the presence of at least one high-memorability noun is sufficient to maintain recognition performance

4.2 Effect of Voice on CMS:

4.2.1 Hypothesis Statement

H_0 : There is no significant difference in CMS between passive-voice and active-voice sentences.

H_a : Passive-voice sentences will yield significantly lower CMS than active-voice sentences.

4.2.2 Rationale

Passive sentences may impose greater syntactic complexity and additional parsing demands during comprehension (Ferreira, 2003), which could reduce encoding efficiency and subsequent recognition memory. However, the effect of passive syntax may operate primarily at the point of comprehension rather than at the level of long-term memory storage.

4.2.3 Statistical Test

A paired Wilcoxon Signed-Rank test was applied to participant-level mean CMS values (Active vs Passive), comparing within-subject differences. This is the methodologically correct approach since each participant contributes both an Active and a Passive CMS score.

4.2.4 Results

No significant difference was found between Active and Passive voice conditions:

W = 1938.5, p = .344, rank-biserial r = .11 (small effect)

Descriptively, Passive sentences showed a slightly higher mean CMS (M=0.716) than Active sentences (M=0.707), opposite to the alternative hypothesis. The effect size r = .11 indicates negligible practical significance.

4.2.5 Visualisation

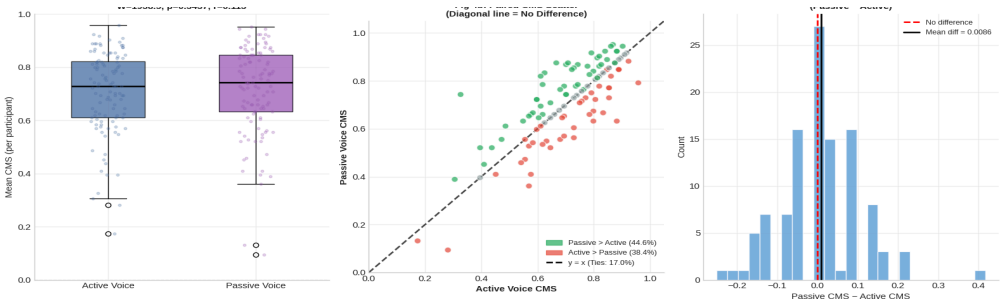


Fig: CMS by Voice Box Plot; Paired CMS Scatter plot (diagonal = No Difference); Distribution of Paired Differences

4.2.6 Conclusion:

H_0 is retained. Voice does not significantly modulate recognition accuracy (CMS). This is consistent with Ferreira (2003), who demonstrated that passive syntactic costs operate at comprehension rather than long-term memory encoding. Importantly, although voice does not affect accuracy, it does affect retrieval speed (H4 - 4.4 Section)

4.3 Effect of Word Memorability on WR Accuracy

4.3.1 Hypothesis Statement

H_0 : There is no significant difference in WR accuracy across the four word memorability conditions.
H_a : Word memorability type significantly affects WR accuracy, such that LL sentences will show lower controlled recognition accuracy than HH, HL, and LH sentences.

4.3.2 Statistical Test & Results

A Friedman test was conducted on participant-level WR accuracy data. The test did not reveal a significant effect of word type on WR accuracy: $\chi^2(3, N=112) = 3.34$, $p = .343$, Kendall's $W = .01$

Descriptively, mean WR accuracy was highest for LH (0.762) and lowest for LL (0.728), consistent with the CMS pattern. However, the differences were small (range ≈ 0.034) and not statistically reliable. Post-hoc tests were not conducted, as the omnibus test was non-significant.

4.3.3 Visualisation

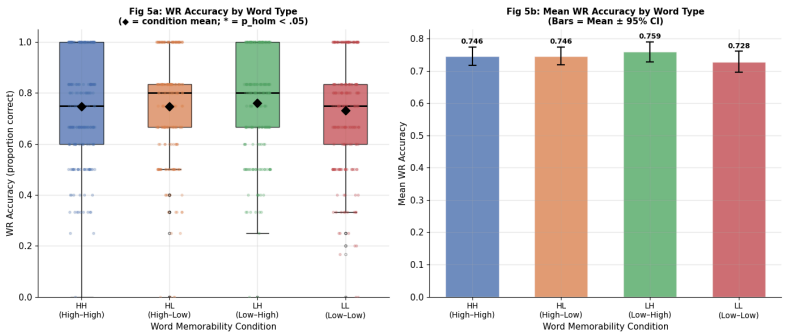


Fig: [left] Box plot of WR accuracy by Word Type; [Right] Mean WR Accuracy by word type

4.3.4 Conclusion

H_0 is retained. WR accuracy did not differ significantly across word memorability conditions. The descriptive pattern (LL lowest: M=0.728) is consistent with the CMS finding in H1 but did not reach statistical significance at N=112. This likely reflects a power limitation: the WR accuracy range across conditions is narrow (≈0.034), requiring a larger sample to detect. The convergence of CMS (H1) and WR accuracy (H3) provides cross-measure support for the LL memorability deficit.

4.4 Voice Affects Retrieval Speed but Not Accuracy

4.4.1 Hypothesis Statement

H_0 : There is no significant difference in IR Reaction Time (RT) between Active and Passive voice.
H_a : Passive voice sentences produce significantly longer IR RT than Active voice sentences, reflecting additional syntactic processing demands at retrieval.

4.4.2 Rationale

Although H2 showed voice does not affect accuracy, equivalent accuracy does not imply equivalent processing efficiency. A sentence may be recognised just as often but require more time. Passive

constructions involve agent-patient reversal, which may slow recognition decisions through additional syntactic verification steps at retrieval.

4.4.3 Statistical Test & Results

Only Hit trials (correct recognitions with IR response) were used, ensuring clean latency estimates. Median IR RT was computed per participant per voice condition, then compared with a paired Wilcoxon Signed-Rank test.Passive sentences were recognised significantly more slowly than active sentences:

$$W = 2193.0,\ p = .0022,\ \text{rank-biserial } r = .331 \text{ (medium effect)}$$

The median paired difference was +44 ms (Passive − Active), indicating that passive voice consistently imposed a retrieval cost.

4.4.4 Visualisation

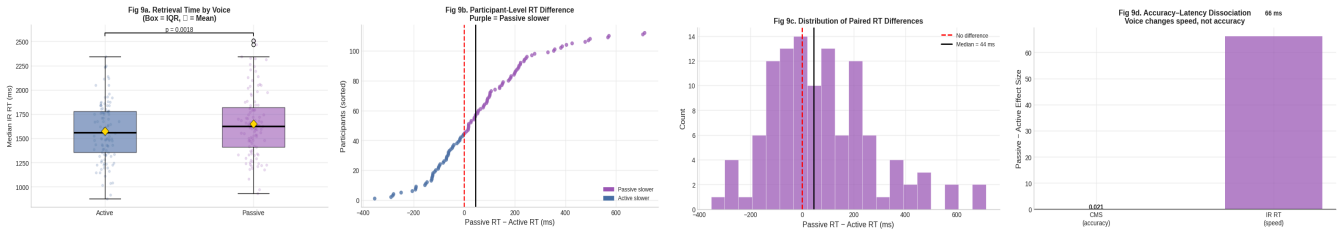


Fig: 1. RT by Voice 2. Participant Level RT differences 3. Distribution of paired RT 4. Accuracy-Latency Dissociation

4.4.5 Conclusion:

H_0 is rejected. Sentence voice significantly affects retrieval latency (p=.002, r=.33) but not retrieval accuracy (H2: p=.344). This accuracy-latency dissociation reveals that passive voice imposes a processing cost specifically at the retrieval stage - participants remember passive sentences just as accurately but require additional time to confirm recognition. This is consistent with psycholinguistic accounts: passive constructions require syntactic role re-mapping (agent appears post-verbally or in a by-phrase), leading to extra verification steps during recognition. Memory strength is preserved; retrieval efficiency is not.

4.5 Trial Level Reaction Time Predicts Recognition Accuracy

4.5.1 Hypothesis Statement

H_0 (Null): There is no significant association between reaction time and accuracy in recognition tasks.
H_a (Alternative): Lower reaction times are associated with lower accuracy in recognition tasks (classic speed-accuracy tradeoff).

4.5.2 Data & Descriptive Approaches

H5 uses the trial-level dataset (reactions_data.csv, ~15,170 trials). Because the RT-accuracy relationship is not obvious from raw counts, three complementary descriptive approaches were applied before inferential testing, each asking the same question from a different angle:

Approach 1 : Median Split (within-participant):
Each participant's own median RT was computed, and their trials were labelled Fast (≤ median) or Slow (> median). Mean accuracy was then compared across these two groups. Using the within-participant median - ensures the split reflects whether a given trial was fast or slow relative to that specific person's baseline, not an arbitrary global threshold. Result: Fast trials averaged 86.92% accuracy vs Slow trials at 62.34% - a gap of over 24 percentage points.

Approach 2 : Within-Participant Z-Score Analysis:
Each trial's RT was standardised by subtracting that participant's mean RT and dividing by their standard deviation. This removes both between-participant speed differences and baseline variability, isolating trial-level fluctuations. Accuracy was compared for trials where $Z < 0$ (faster than own average) vs $Z > 0$ (slower). Result: $Z < 0$ trials had mean accuracy 86.52%, $Z > 0$ trials had 62.19%. Mean z-score for correct trials = −0.2217; for incorrect = +0.5859, confirming correct responses are systematically faster relative to each person's own rhythm.

Approach 3 : Pearson Correlation (overall and per-participant):
A global Pearson correlation between raw RT and binary accuracy (coded 0/1) was computed as a continuous measure of association across all trials. Then the same correlation was computed separately for each participant to check consistency. Result: Overall $r = -0.3421$ ($p < .001$). Per-participant mean r =

−0.3746; 113 of 114 participants individually showed a negative correlation, confirming the effect is not driven by a minority of outliers.

Data Representation	Test Applied
Median-split groups (Fast vs Slow)	Mann-Whitney U
Same median-split as 2×2 contingency table	Chi-Square (Cramér's V)
Continuous raw RT vs binary accuracy	Spearman Correlation
Within-participant z-scored RT → accuracy (trial-level)	Naive Logistic Regression
Same z-scored RT, grouped by PID (clustering-corrected)	GEE Logistic Regression
Same z-scored RT with random intercepts per PID	Bayesian GLMM (variational)

4.5.3 Inferential Statistics

All six approaches yield a consistent and significant NEGATIVE relationship between RT and accuracy

The table below summarises all results:

Test	Metric	Value	p-value	Interpretation
Logistic Regression	RT_Z Coefficient	−0.8120	<.001	Faster = More Accurate
GEE (clustering-corrected)	RT_Z Coefficient	−0.7981	<.001	Significant after clustering correction
Bayesian GLMM	RT_Z Coefficient	−0.9021	(Bayesian)	Consistent with GEE
Spearman Correlation	Rho (ρ)	−0.3287	<.001	Significant negative correlation
Mann-Whitney U	U-Statistic	3.6×10^{7}	<.001	Fast=85.5% accurate; Slow=59.4%
Chi-Square	Cramér's V	0.2927	<.001	Small-to-moderate effect

The GEE model (the primary and most rigorous test) produces a coefficient of −0.7981 (OR = 0.4502, 95% CI [0.4227, 0.4794], $p = 1.96\times10^{-136}$). This means each 1 SD increase in RT (i.e., being 1 SD slower than one's own average) reduces the odds of a correct response by approximately 55%. Crucially, the GEE confidence interval lies entirely below 1.0, ruling out any null or positive effect even after accounting for within-participant clustering.

Chi-Square provides complementary confirmation without the tied-rank concerns of Mann-Whitney: $\chi^2(1)=1299.69$, Cramér's V=0.29, confirming a small-to-moderate practical effect at the group level. The Bayesian GLMM (coefficient=−0.9021) is consistent with GEE, with the larger magnitude reflecting the shift from population-average to subject-specific estimation.

4.5.4 Visualizations

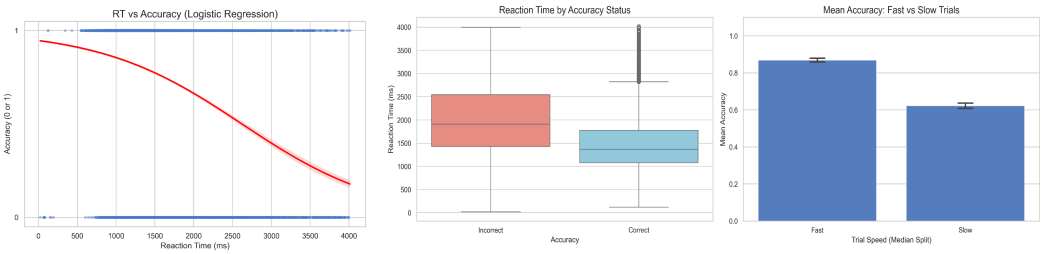


Fig: 1. RTvs Accuracy Logistic Regression; 2. Reaction time Box Plot; 3. Mean Accuracy of Fast and Slow trials

4.5.6 Conclusion:

H_0 is rejected but in the opposite direction to the prediction: faster responses are significantly more accurate, not less. This reversed speed-accuracy pattern is consistent across all six analytical approaches and is robust across 112 participants individually. Rather than a speed-accuracy tradeoff, the data reveal a decisiveness effect: when memory strength is high, participants recognise the sentence quickly and correctly; when it is low, they hesitate, respond slowly, and err. The GEE analysis confirms this is a genuine population-level effect and not a statistical artefact of treating correlated trials as independent observations.

4.6 Cognitive Fatigue Across Experimental Blocks

4.6.1 Hypothesis Statement

H_0 : There is no significant change in accuracy or RT across blocks.

H_a (Alternative): Recognition accuracy decreases and reaction time increases across blocks due to cognitive fatigue.

4.6.2 Data Pipeline & Variables

uses block_accuracy_data.csv - the block-level aggregate dataset. Each row represents one participant's performance summary for one block. Two variables are analysed: Reaction Accuracy per block and Mean RT per block. Participant-level trend analysis was also conducted: for each participant, Block 3 values were compared to Block 1 to classify RT as Inc. or Dec., and accuracy as Inc. or Dec.

4.6.3 Descriptive Statistics:

Block	Mean RT (ms)	Median RT (ms)	Mean Accuracy	RT Change	Acc Change
1	1688.25	1612	67.93%	—	—
2	1632.43	1558	77.16%	improved	improved
3	1602.52	1532	82.73%	improved	improved

4.6.4 Inferential Statistics:

Friedman tests assessed block differences in RT and accuracy across the 3 within-subject blocks, showing significant effects for RT, $\chi^2(2) = 9.81$, $p = 7.42 \times 10^{-3}$, Kendall's W = .043, and accuracy, $\chi^2(2) = 88.48$, $p = 6.12 \times 10^{-20}$, Kendall's W = .388. Second, significant omnibus results were followed by pairwise Wilcoxon Signed-Rank tests for Block 1 vs 2, Block 2 vs 3, and Block 1 vs 3. Third, Bonferroni correction controlled family-wise error across comparisons, giving an adjusted threshold of $\alpha_corrected = 0.0167$ (0.05/3).

Comparison	RT p-value	RT Direction	Accuracy p-value	Acc Direction
B1 vs B2	8.59×10^{-3}	improved	1.16×10^{-11}	improved
B2 vs B3	1.31×10^{-1}	—	4.44×10^{-7}	improved
B1 vs B3	2.09×10^{-3}	improved	2.65×10^{-16}	improved

4.6.4 Visualizations

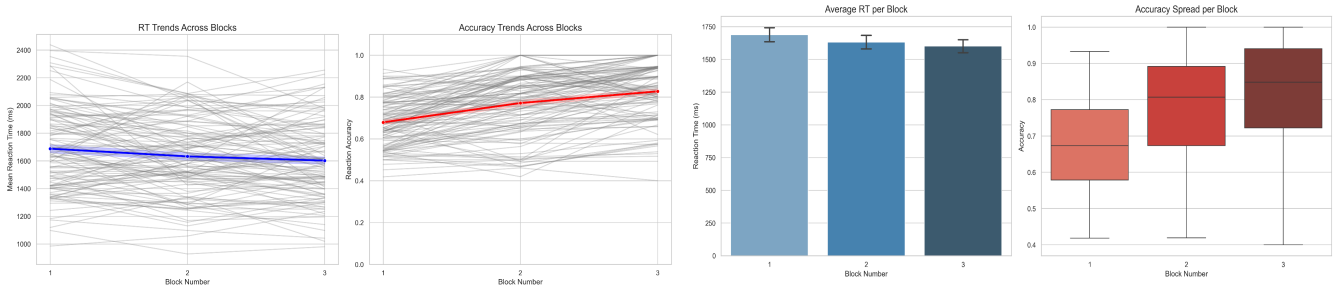


Fig: Block level Accuracy trends (left 2) spaghetti plot, Bar graph and Box plots (right most)

4.6.5 Conclusion:

H_0 is rejected - but the direction of change directly contradicts H6. Significant block-level changes exist in both RT and accuracy, but both improved rather than declined. Accuracy improved strongly across all three block transitions (all Bonferroni-corrected $p < .001$), while RT improvement was concentrated in the first transition (Block 1→2: p=.009) and then plateaued (Block 2 to 3: p=.131, not significant). The moderate effect on accuracy (Kendall's W=.39) vs negligible effect on RT (W=.04) indicates that response quality benefited substantially from practice, while speed gains were smaller and front-loaded. This pattern - rapid early improvement that then levels off - is the hallmark of a learning curve, not a fatigue effect. Future research should use counterbalanced block order and extended paradigms (more blocks) to separate learning from fatigue at the individual participant level.

4.7 Effect of Syntactic Normalization on WR Accuracy

4.7.1 Hypothesis Statement:

[H_0]: There is no significant difference in word-recognition accuracy between sentences repeated in the same voice and sentences where the voice has been transformed.

[H_a]: Word-recognition accuracy is significantly lower for transformed sentences than for same-voice sentences.

4.7.2 Rationale:

While participants effectively recognize the general occurrence of a sentence, memory representations are biased toward underlying semantic meaning. Surface-level syntactic details (such as whether a sentence was active or passive) are hypothesized to be poorly maintained.

4.7.3 Statistical Test & Results:
A paired Wilcoxon Signed-Rank test was performed on per-participant mean Word Recognition (WR) proportions.
Descriptive: Participants correctly identified the wording 80.95% of the time when the voice remained unchanged. When the syntax was transformed (Active ↔ Passive), accuracy dropped to 68.15%, revealing a clear 12.80% accuracy penalty.
Inferential: The accuracy gap was highly significant ($W = 4521.0$, $p = 6.59 \times 10^{-7}$). The effect size (rank-biserial $r = 0.536$) is classified as a "Large" magnitude effect.

4.7.4 Visualizations:

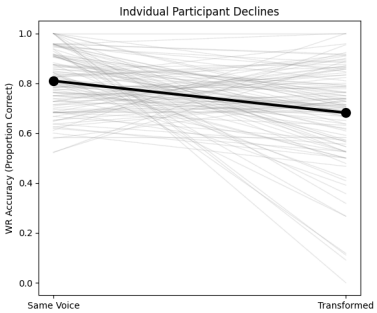


Fig: Individual participant accuracy changes across Same vs. Transformed conditions (black line is group mean).

4.7.5 Conclusion:
H_0 is rejected. Syntactic transformation of a sentence's voice causes a substantial, consistent negative impact on a participant's ability to identify the original wording.

4.8 Memory Strength Model
4.8.1 Hypothesis Statement:
[H_0]: Item-recognition latency (RT_IR) does NOT predict the accuracy of subsequent wording retrieval.
[H_a]: Faster IR reaction times predict HIGHER WR accuracy.

4.8.2 Rationale:
The 'Memory Strength' account suggests that stronger memory traces are both faster to access and more detailed. Rather than testing if "fast people are more accurate," this tests if when a person is faster than their own average, they are more likely to retrieve the exact wording correctly.

4.8.3 Statistical Test & Results:
Using Generalized Estimating Equations (GEE) with a Binomial family, continuous true within-subject z-scored RTs (RT_within_z) were used to predict binary WR accuracy.

Descriptives: Participants' fastest relative quartile of responses (Q1) yielded 78.28% accuracy, which steadily declined to 72.08% for their slowest relative quartile (Q4).
Primary Model: The unadjusted model showed a significant negative association ($\beta = -0.1177$, $SE = 0.0341$, $p = 0.0006$; $OR = 0.8890$).
Adjusted Model: When task difficulty (Condition and Voice) was added as a covariate, the RT effect was attenuated to non-significance ($\beta = -0.0610$, $p = 0.0875$; $OR = 0.9408$).

4.8.4 Visualizations:

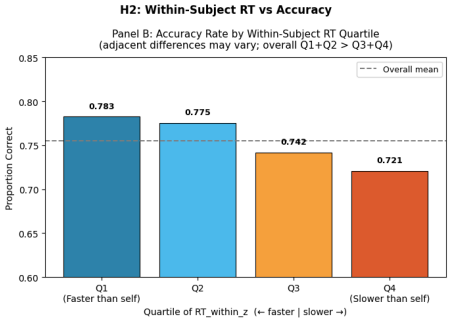


Fig: Proportion of correct wording-retrieval responses across quartiles of within-subject standardized reaction times.

4.8.5 Conclusion:
H_0 is rejected for the primary model. When participants recognize an item faster than their own baseline, they are significantly more likely to correctly remember its specific wording, supporting the memory-strength relationship. However, the attenuation of this effect in the adjusted model indicates that part of this speed-accuracy correlation is driven by the structural difficulty of the sentences themselves (shared variance).

Final Conclusion

This study investigated sentence memorability across 112 participants using a continuous recognition paradigm. Key findings are as follows:

Word memorability type significantly influenced recognition, with LL sentences consistently scoring lower than all other conditions, confirming that even a single high-memorability noun is sufficient to sustain recognition performance. Grammatical voice, however, did not affect recognition accuracy but did impose a measurable retrieval cost, with passive sentences taking significantly longer to recognise. WR accuracy showed a descriptively consistent but statistically non-significant trend across memorability conditions, likely reflecting a power limitation.

Contrary to the classic speed-accuracy tradeoff, faster responses were significantly more accurate, pointing to a decisiveness effect driven by memory strength rather than a speed-accuracy exchange. Performance improved rather than declined across experimental blocks, indicating a learning effect. Syntactic transformation of voice produced a substantial accuracy penalty in wording retrieval, confirming that surface-level syntactic form is retained in memory. Finally, within-subject RT fluctuations predicted wording retrieval accuracy in the unadjusted model, though this effect was partially explained by sentence-level difficulty.

Overall, sentence memorability is primarily shaped by noun-level memorability, while grammatical voice influences retrieval efficiency rather than recognition accuracy. These results support Dual Coding Theory and psycholinguistic accounts of passive processing, while also revealing a memory-strength-driven decisiveness effect in recognition behaviour.

Contributions

Vishnu Varun
Data preparation and processing; block validation; Hypotheses H1–H4 analysis and visualisation; descriptive statistics (WR section); normality testing.
Pavan Karke
Introduction; hypothesis formulation; experimental design and variables; Hypotheses H7–H8 analysis and visualisation; conclusion.
Susheel Krishna
Descriptive statistics (Memorability by Voice and interaction exploration); Hypotheses H5–H6 analysis and visualisation; Shapiro–Wilk normality analysis; inferential statistics for H3.

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